OPEN PLATFORM FIRMWARE (OPF)



Decoupling Silicon Dependencies from Control-Plane Firmware

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Hyperscaler Heterogeneous Fleet

Every axis below multiplies the management-firmware matrix.

Multiple architectures in service	x86, Arm and GPU/accelerator SoCs run side by side, each with its own management interfaces and RAS model.
Multiple management controller suppliers	Management SoCs come from multiple vendors, so silicon-specific code must work across every BMC platform in the fleet.
Several silicon generations at once	A generation remains in service for years after its successor arrives, so integration work repeats across generations.
Need to support the fleet for 7+ years	Fleet issue mitigations and solutions must scale and remain serviceable for years.

Every axis multiplies the number of firmware images we have to build, test and deploy.

What breaks today

Custom Interfaces for Silicon Vendors	CPUs and GPUs still use custom interfaces to communicate with the control plane.
Vendor-Specific Silicon Patches	Vendor patches in control-plane firmware support custom silicon interfaces and features.
Silicon-specific services	Silicon telemetry, RAS, crash dump, debug and recovery features, power policies, and GPU interconnect links require custom BMC changes for each vendor and SKU.
Version-matched deployments	A silicon mitigation can require microcode, host firmware, and BMC firmware to move together.

Each of these lands as vendor-specific code in the firmware image.

What It Costs

Per Generation

Integration and qualification repeat for every silicon vendor and often for every generation.

Maintenance

Silicon-specific management SoC code increases maintenance complexity at fleet scale.

Days to Months

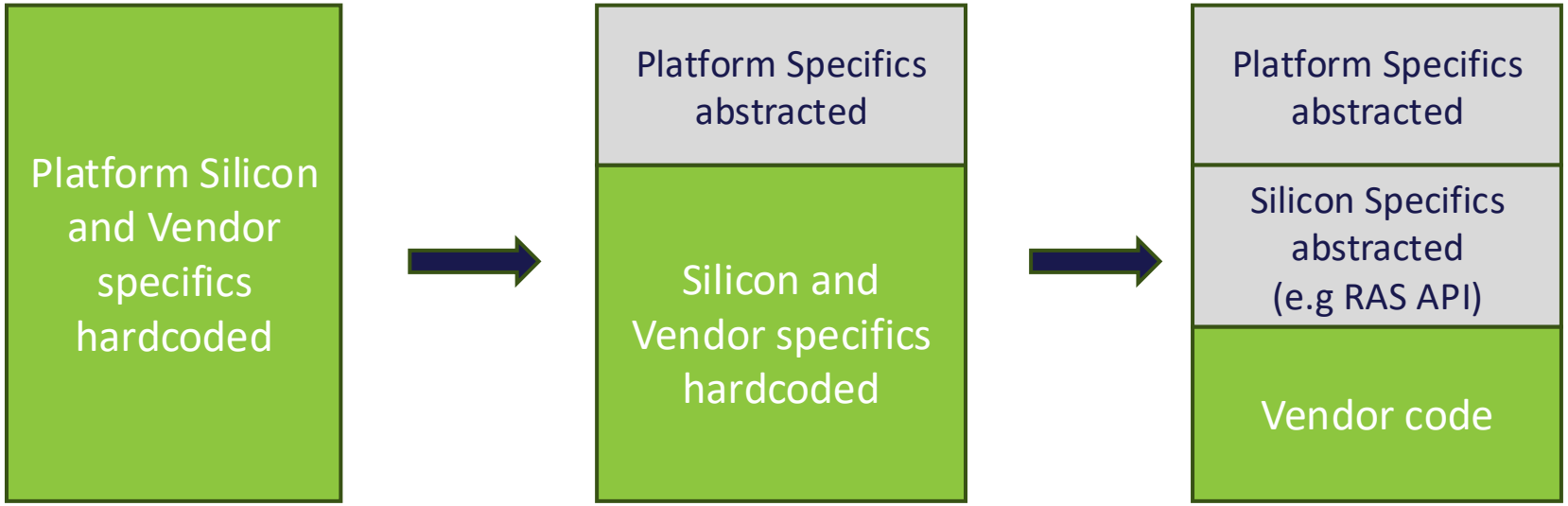
Worst-case time from silicon-fix availability to deployment in the fleet

- **Configuration Fix vs. Code Fix**

An incorrect register, bus, or header value in configuration can be addressed with a file push and service restart; a code update requires a full firmware release.

What you must change defines the cost. A configuration change is a file push; a new image requires a fleet rollout.

The ‘Portability’ Arc



Options for integrating Vendor-specific code

OPTION	WHAT IT BUYS	WHAT IT COSTS
Extend the schemas until data suffices	Auditable. Declarative. Builds on existing abstractions.	Cannot express all silicon-specific behavior. The algorithmic tail eventually escapes the schema.
Bind vendor code into the BMC image	Deterministic. One validated artifact. Fits today's build and qualification model.	Silicon-specific changes and fixes remain coupled to the BMC lifecycle — rebuild, requalify, and roll out the image.
Bind Vendor Components independently	Decouples silicon and BMC release cycles. Silicon-specific changes and fixes can be serviced independently.	Requires a stable interface and a trust, versioning, lifecycle, and isolation model.

Can data abstraction take us far enough? If not, does the remaining executable silicon-specific logic bind at build time or later?

Statically linked vs late binding

The same inputs in both lanes. The only difference is where they are bound.

TODAY – EVERYTHING BINDS AT BUILD



Change any one input and the whole chain reruns — rebuild, requalify, reflash every node.

IF VENDOR COMPONENTS COULD BIND LATER



Fixes and enhancements are a component push and a service restart — not a rebuild, requalification and fleet reflash.

Late-binding Requirements

Stable late-binding interface	Vendor-neutral contract between the BMC and independently delivered components.
Vendor-signed, BMC-verified	Authenticity from the silicon vendor's signature, checked against a key anchored in the BMC's own root of trust.
Measured and attestable	The BMC measures each component as it loads, so the fleet can verify exactly what is running on each node.
Anti-rollback and revocation	The operator can prevent rollback to vulnerable versions and revoke compromised components or signing keys.
Owner Control	The platform owner retains final authority over which vendors and components are permitted to execute.

Call to Action

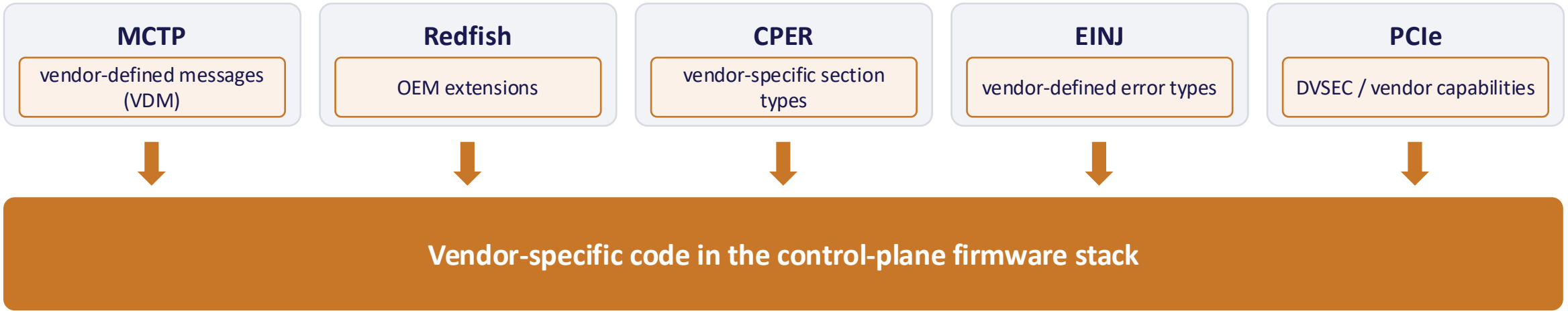
Feedback through OSF workstreams

Management SoC firmware is statically bound to SoC-specific vendor libraries — one per silicon vendor, often one per generation. Every stepping, errata fix or policy change forces a BMC image rebuild, requalification and a coordinated fleet rollout.

Thank You!



Vendor Specific extensions



- CADENCE — SILICON SHIPS FASTER THAN SPECS EVOLVE
- DIFFERENTIATION — BEHAVIOUR TIED TO THE VENDOR'S OWN IP

Leads to Vendor code in the control plane